

After wasting 8 years of life, if enough efforts are given in 1 year, can someone reach target/reachable target. Using a dynamical model that includes frontier drift, receding penalty-guilt feedback loop and financial constraint.

⇒ State vector, $x(t) = \begin{pmatrix} A(t) \\ R(t) \\ S(t) \\ G(t) \end{pmatrix}$

A = Absolute legible output (unit)

R = External Frontier (unit)

G = Guilt/hopelessness (≥ 0)

Derived quantity, $c(t) = A(t) - R(t)$ [$c < 0$ means below parity]

c is not independent state, it's linear function of two states.

Control u_A = hours per year directed as producing legible output.

Constraint set, as hours bound by duty/cycle:

$$0 \leq u_A \leq H n(G) (1 - \omega) \quad n(G) = n_0 e^{-\mu G}$$

n → function of available days usable
 ω → fraction of hours produced into work

$$\dot{A} = \gamma \pi$$

Active Capital → $\dot{A} = \gamma \pi \phi(S) u_A - \delta_A A$

Shared Resource → $\dot{R} = \zeta + g_R R$

Structural Support → $\dot{S} = p_s n u_S + p_1 (1 - \pi) u_A - \delta_S S$

Constraint Factor → $\dot{G} = g_0 (1 - n e^{-\mu G}) + g_e (1 - \hat{E}) - g_c G$

The constraint eqⁿ:

$$u_A + u_S \leq L(G) := H (1 - \omega) n_0 e^{-\mu G}$$

$\downarrow$ Central Input $\downarrow$ Appears in budget and set by houses

total available energy can not exceed $L(G)$. As G total system capacity to invest in A or S is

change rate in stored Reserve

$$\dot{R} = \gamma + g_R R \rightarrow \text{first order nonhomogeneous ODE}$$

$\dot{R}$ contain no control and no other state, so it's decoupled and solvable alone with approximation.

$$\begin{aligned} \frac{d}{dt} [R e^{-g_R t}] &= \dot{R} e^{-g_R t} - g_R R e^{-g_R t} \\ &= (\dot{R} - g_R R) e^{-g_R t} \\ &= \gamma e^{-g_R t} \end{aligned}$$

At the moment effort year begins:

$$\begin{aligned} R_0 &= R(0) \\ &= \frac{\gamma}{g_R} (e^{g_R h} - 1) \end{aligned}$$

Here, $\phi(s) = \frac{s}{s + s_{1/2}}$, $\Xi(d) = \frac{d}{d + d_{en}}$, $k(c) = 1 - \theta \Xi(d)$, $d := (-c)_+$

$y_+ \rightarrow \max(y, 0)$. Both ϕ, Ξ rise, saturate and concave, $\theta \in [0, 1]$ so k never hits zero, $\hat{E}$ rises with c .

Capability:

$$b = p s_{1/2} v_s + A(1 - \pi v) v_A$$

$$s(s) = s_{\infty} + (s_i - s_{\infty}) e^{-s s} s$$

$$\Rightarrow s_0 = s_{\infty} (1 - e^{-s h}) + s_i e^{-s h}$$

$h \rightarrow$ duration of time passed
 $s_i \rightarrow$ initial capability at the start
 $s_0 \rightarrow$ present day capability
 $s \rightarrow$ change in capability
 $b \rightarrow$ constant source of new capability
 $s_{\infty} \rightarrow$ Absolute maximum capability

A weighted average of where the student started and where it's heading, weight set by $s h$.

$S_h \rightarrow$ controls how fast the transition happens

Fraction of $\frac{\text{Indirect Growth}}{\text{Total Growth}} = \frac{A}{A_1} = \frac{p_I(1-\pi)u_A}{p_S \eta_h u_S + p_I(1-\pi)u_A}$

Output:

$$\frac{dA}{dt} = \frac{dA}{ds} = \beta \phi(s(s) - sA) \quad A(0) = 0$$

Integrating factor e^{sA} ,

$$A_0 = \beta e^{-sA} \int_0^h \phi(s(s)) e^{sA} ds$$

Weighted mean value theorem gives us:

$$A_0 = \frac{\beta \phi}{sA} (1 - e^{-sA}) \quad \bar{\phi} \in [\phi(s_1), \phi(s_0)]$$

Here, $A_0 \leq \beta \bar{\phi} / sA \rightarrow$ bounded

Deficit in knowledge base of the field:

The gap with the student and the total knowledge base required for the field will keep on increasing for 2 reasons: 1. Old knowledge gets irrelevant
2. The field keeps advancing

$$L_0 = R_0 - A_0$$

deficit in the moment effort begin

$$d_0 = R_0 - A_0 = \frac{\gamma}{g_R} (e^{\gamma h} - 1) - \frac{B\bar{\phi}}{s_A} (1 - e^{-s_A h})$$

$\bar{\phi} < 1$ always, and $d_0 \rightarrow \infty$ as $h \rightarrow \infty$

hence

$$\begin{aligned} s_A > 0 \text{ (any } g_R > 0) &\rightarrow A_0 \text{ bounded} \rightarrow R_0 \text{ unbounded} \rightarrow d_0 \rightarrow \infty \\ s_A = 0 \text{ (} g_R > 0) &\rightarrow A_0 \text{ linear in } h \rightarrow R_0 \text{ exponential} \rightarrow d_0 \rightarrow \infty \\ s_A = 0 \text{ (} g_R = 0) &\rightarrow B\bar{\phi}h \rightarrow R_0 = \gamma h \rightarrow (\gamma - B\bar{\phi})h \end{aligned}$$

$$d_0 \rightarrow \infty \iff s_A > 0 \text{ or } g_R > 0$$

$\gamma \rightarrow$ field advancement rate

$g_R \rightarrow$ how much of the standard of growth feeds into work

$h \rightarrow$ length of history in years

$e^{\gamma h} \rightarrow$ How much bigger the drift becomes after h years

$\bar{\phi} \rightarrow$ Average capability multiplier

$s_A \rightarrow$ Rate of which old output ages decays

Effort Year: $\dot{c} = \dot{A} - \dot{R}$
 $= \alpha k(c) e^{-\mu c} - s_A A - \gamma - g_R R$

$A = cR$. So, $\dot{c} = \alpha k(c) e^{-\mu c} - s_A c - \gamma_{eff}$

$$\alpha \equiv \delta \pi \phi_0 H (1 - \omega) n_0, \quad \gamma_{eff} = \gamma + (g_R + s_A) R_0$$

$$\gamma_{eff} = \gamma \left[1 + \left(1 + \frac{S_A}{\theta_R} \right) (e^{\theta_R h} - 1) \right] - (1 - \gamma) \frac{\gamma}{\theta_R} - \gamma A - \gamma = 0$$

if

effective drug

Critical History length:

$$TT := \frac{a}{\gamma_{eff}} \geq 1$$

If $TT < 1$ then $\dot{c} < 0 \rightarrow$ Maximum effort, zero guilt till capacity

$TT_0 := a/\gamma$, the ratio with no history

$$TT(h) = \frac{TT_0}{1 + \left(1 + S_A/\theta_R \right) (e^{\theta_R h} - 1)}$$

Set $TT(h_c) = 1$:

$$(e^{\theta_R h_c} - 1) = \frac{TT_0 - 1}{1 + S_A/\theta_R} = \frac{(TT_0 - 1)\theta_R}{\theta_R + S_A} \quad \text{with } TT_0 < 1$$

$$h_c = \frac{1}{\theta_R} \ln \left[1 + \frac{(TT_0 - 1)\theta_R}{\theta_R + S_A} \right]$$

Critical History length

Linearizing

$$\dot{x} = f(x)$$

$$f(x^*) = 0$$

$$x = x^* + \delta \longrightarrow \dot{x}^* \text{ is constant}$$

$$\Rightarrow \dot{x} = \dot{\delta}$$

As Taylor series: $f(x^* + \delta) = f(x^*) + f'(x^*)\delta + \frac{1}{2}f''(x^*)\delta^2 + \dots$

$$\dot{\delta} = f'(x^*)\delta$$

$$\Rightarrow \delta(t) = \delta_0 e^{f'(x^*)t}$$

using partial derivatives —

$$\dot{\delta}_c = \frac{\partial F}{\partial c} \delta_c + \frac{\partial F}{\partial \mu} \delta_\mu, \quad \dot{\delta}_\mu = \frac{\partial F}{\partial c} \delta_c + \frac{\partial F}{\partial \mu} \delta_\mu$$

$$\dot{\delta} = J\delta$$

$$J_{ij} = \left. \frac{\partial F_i}{\partial x_j} \right|_{x^*}$$

As we have 2 linear equation with four coefficient it can be written with

Jacobian.

$$E := e^{-\mu \kappa^*}$$

$$\eta = \eta_0 E$$

$$\frac{\partial \kappa}{\partial c} = -0 \left[(d) \cdot (-1) \right] = + \frac{\eta_{ch}}{(d + \eta)^2}$$

$$J_{11} = \frac{\eta E \eta_{ch}}{(d + \eta)^2} - S_A, \quad J_{12} = -\eta \mu \kappa E, \quad J_{21} = -\eta E'(\eta)$$

$$J_{22} = -\beta$$

Matrix form:

$$J = \begin{pmatrix} \frac{\eta E \eta_{ch}}{(d + \eta)^2} - S_A & -\eta \mu \kappa E \\ -\eta E'(\eta) & -\beta \end{pmatrix}$$

with $P_g := A_K - g_w N K$, the net guilt damping

$$J = \begin{bmatrix} \frac{aEOch}{(d+ch)^2} - S_A & -\alpha_k K E \\ -g_c \hat{E}' & P_g \end{bmatrix}$$

$$\dot{S}_c = \left[\frac{aEOch}{(d+ch)^2} - S_A \right] - \alpha_k K E S_c$$

$$\dot{S}_c = -g_c \hat{E}' S_c - P_g S_c$$

Here $J_{12} < 0 \rightarrow$ Guilt cause drops, costs output

$J_{21} < 0 \rightarrow$ falling behind brings guilt

$J_{22} < 0 \rightarrow$ when guilt is self limiting

Saddle Point:

$$\det J = -P_g \left[\frac{aEOch}{(d+ch)^2} - S_A \right] - \alpha_k K E g_c \hat{E}'$$

here,

$$\text{Saddle} \Leftrightarrow \frac{aEOch}{(d+ch)^2} + \frac{\alpha_k K E g_c \hat{E}'}{P_g} > S_A$$

Rentry Feedback

Guilt Loop

A_g

$$d_{\text{stable}} \sim \sqrt{\frac{aEOch}{SA}} - Ch$$

Something Efforts can not touch:

$$\frac{d}{dt} \begin{pmatrix} A \\ R \end{pmatrix} = \begin{pmatrix} -SA & 0 \\ 0 & g_R \end{pmatrix} \begin{pmatrix} A \\ R \end{pmatrix} + \begin{pmatrix} b \\ 0 \end{pmatrix} V_A + \begin{pmatrix} 0 \\ y \end{pmatrix}, \quad b := \sigma \pi \phi_0$$

$$\dot{A} = -SA + bV_A$$

$$\dot{R} = g_R R + y$$

$$x(t) = \int_0^T e^{A(T-\tau)} B u d\tau, \text{ by expanding,}$$

$$x(t) = \sum_{k=0}^{\infty} A^k B C_k$$

$$C_k = \int_0^T \frac{(T-\tau)^k}{k!} U(\tau) d\tau$$

$$C = [B \ AB] = \begin{pmatrix} b & -SAb \\ 0 & 0 \end{pmatrix} \quad \text{rank } 1 < 2$$

The best Possible Horizon:

$\dot{A} = -SA + bV$, the Generation is:

$$W(t) = \int_0^T e^{-2SA(T-\tau)} b^2 d\tau = b^2 \frac{1 - e^{-2SAT}}{2SA}$$

$$V^*(t) = b e^{-SA(T-\tau)} W^{-1} \Delta$$

Integrating and simplifying with $1 - e^{-2x} = (1 - e^{-x})(1 + e^{-x})$:

$$H(T) = \frac{2\Delta(T)}{b(1+e^{-S_A T})}, \quad \Delta(T) = A_{\text{tgt}}(T) - e^{-S_A T} A_0$$

Against the budget LT ,

$$R(T) = \frac{2\Delta T}{bLT(1+e^{-S_A T})}, \quad \text{feasible} \Leftrightarrow R(T) \leq 1$$

Positivity Target, $A_{\text{tgt}}(T) = R(T)$ and for T large enough $S_A T \gg 1$

$$R(T) \sim \frac{2\gamma e^{\theta R h}}{\theta_R bL} \cdot \frac{e^{\theta R T}}{T}$$

Since, $\frac{d}{dT} \frac{e^{\theta R T}}{T} = \frac{e^{\theta R T}(\theta_R T - 1)}{T^2}$, it vanishes at

$$T^* = \frac{1}{\theta_R}$$

Positivity reachable at some horizon $\Leftrightarrow R(T^*) \leq 1$

A_{tgt} fixed, Δ is bounded and $R \sim 2A_{\text{tgt}} / bLT \rightarrow 0$

When, $H = LT$:

$$\Delta_{\text{max}}(T) = \frac{bLT(1+e^{-S_A T})}{2}$$

$S_A T \rightarrow 0$ becomes bLT

Summary of this:

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It'll be impossible for ~~me to~~ for the student to ever match the output to their input. But as long as they continue to go on without any guilt, they ~~can~~ they can work the best that day, that's a great ~~of~~ output toward that goal.

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1. (10) (10) (10) (10) (10) - 1

2. (10) (10) (10) (10) (10) - 2

3. (10) (10) (10) (10) (10) - 3

4. (10) (10) (10) (10) (10) - 4

5. (10) (10) (10) (10) (10) - 5

6. (10) (10) (10) (10) (10) - 6

7. (10) (10) (10) (10) (10) - 7

8. (10) (10) (10) (10) (10) - 8

9. (10) (10) (10) (10) (10) - 9

10. (10) (10) (10) (10) (10) - 10